



National University of Computer & Emerging Sciences, Karachi

Fall-2017 CS-Department

MidTerm 2

27th October 2017, 9:00 am – 10am



Course Code: CS203	Course Name: Database Systems
Instructor Name / Names: Dr Zeeshan Ahmed, Mr. Ahsan Shah, Miss Tania Iram	
Student Roll No:	Section No:

Instructions:

- Return the question paper.
- Read each question completely before answering it. There are 3 questions and 2 pages.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- This paper is subjective

Time: 60 minutes.

Max Marks: 50 points

Question 1:

Marks:15

Consider the following Constable Training Database schema instances:

Training_passed

Constable_id	Course_title	grade
CB101	Firing	A
CB101	Observation	F
CB102	Firing	A
CB102	Psychology	A
CB102	Physical Fitness	B
CB103	Observation	C
CB103	Psychology	A
CB104	Psychology	B
CB104	Firing	B

Courses

Course_title	Level
Physical Fitness	1
Observation	2
Psychology	3
Firing	4

Costables

Constable_id	C_name
CB101	Ali
CB102	Rafeeq
CB103	Pervaiz
CB104	Tariq

Assumption: Salaried_commission are possible for constables who passed both level 1 and 2 courses.

Write the relational algebra expressions and SQL queries for each enquiry given below:

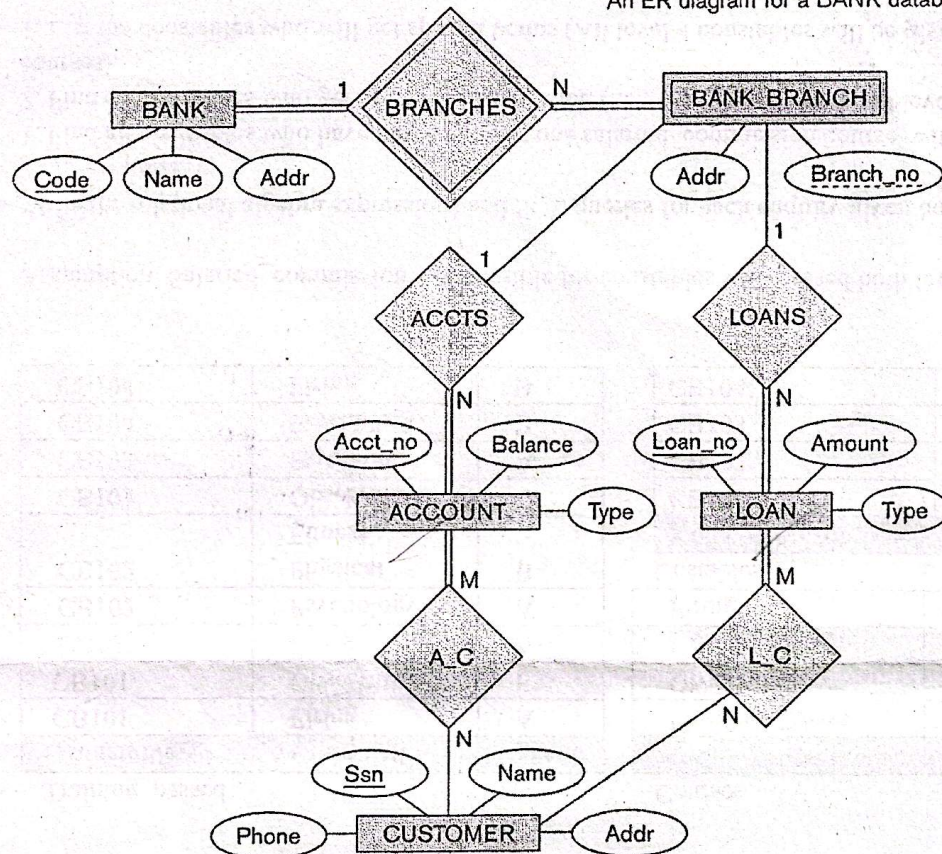
1. Find all constables who have passed atleast one salaried_commission course with 'A' grade.
2. Find all constables who get salaried_commission (i.e., who have passed both level 1 and 2 courses).
3. List the constables who will get special bonus (All level 4 constables will be given special bonus).

Question 2:

Marks:15

Translate the following ER diagram into Relational model using the concepts of ER to relational mapping.

An ER diagram for a BANK database schema.



Question 3:

Marks:20

Consider the relation R with attributes; $R = \{A, B, C, D, E, F, G, H, I, J\}$. Suppose that the following functional dependencies hold on R :

- $\{A\} \rightarrow \{B, C, D, E, F, G, H\}$
- $\{B, C\} \rightarrow \{A, D, E, F, G, H\}$
- $\{E\} \rightarrow \{F\}$
- $\{G\} \rightarrow \{B\}$
- $\{C\} \rightarrow \{H\}$
- $\{H\} \rightarrow \{I, J\}$

Normalize this relation into 1NF, 2NF, 3NF and BCNF.